



**Politecnico
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Lab report for the course Qubit Electronics

Master degree in Quantum Engineering

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1. Introduction

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2. Changes in python script

In order to make the measurements as described in the Introduction section, this time the value of the central barrier gate is swept in a range that goes from 0.37 V to 0.42 V, taking 6 equally spaced values within this range:

```
V_Barrier_Vec_2 = np.linspace(0.37,0.42, 6)
```

For each value of the central barrier gate, the program solves the non-linear Poisson equation and subsequently the Schrodinger equation, in order to obtain the energy levels of the system analyzed. The workflow of the program consists of the following procedure:

- Definition of the materials assigning to each region of the system their corresponding physical properties, as in the previous laboratories.
- Set up of the parameters for solving both the Poisson and the Schrodinger equations.
- Definition of the loop that sweeps the value of the central barrier gate:

```
energy_arr = np.zeros((len(V_Barrier_Vec_2),6))
energy_arr[:,0] = V_Barrier_Vec_2
p = PoissonSolver(d, solver_params=params_poisson)

for idx, V_barrier_2 in enumerate(V_Barrier_Vec_2):
    d.new_gate_bnd('surface_Plunger_R_X_gate', V_Plunger_Right_Vec[idx], mWF)
    d.new_gate_bnd('surface_Barrier_M_X_gate', V_barrier_2, mWF)

    p.solve()
    d.set_V_from_phi()
    submesh = SubMesh(d.mesh, dot_region)
    dot = SubDevice(d, submesh)
    s = SchrodingerSolver(dot, solver_params=params_schrod)
    s.solve()

    energy_arr[idx,1:] = dot.energies/ct.e
    dot.print_energies()
```

3. Analysis of the results

3.1 Physical setting

here explain all the physical concerns of the system, like start from describing the system as 2 quantum systems ψ_{sr} and ψ_{sl} , then write the hamiltonian and describe what are the matrix elements. explain in detail the t element (offdiagonal term) and then introduce the theory of WKB and how it can be used to recover that term. then see from the matrix that the splitting is proportional to t and from WKB that t is proportional to the exponential of the V applied $\rightarrow$ so the splitting is proportional to the exponential of the V applied. Then show the results (next section) and after see the fit and comment on how the fit is close to the experimental results.

3.2 Energy splittings and fitting of the results

In our simulation, the energy splittings between the ground states of the two dots are reported in the following table:

Barrier Voltage (V)	Energy splitting (μeV)
3.69	2.08
3.80	2.91
3.90	3.83
3.99	4.84
4.09	5.94
4.19	7.20

We then fitted the experimental data to an exponential function of the form:

$$y = a \cdot e^{b \cdot x} \quad (3.1)$$

where y in this case represents the energy splitting and x is the barrier voltage. The showed function has been optimized to best fit the experimental data and the parameters a and b have been determined, resulting in $5.11e-10$ and 22.79 respectively.

The plot of the fitted curve and the experimental data is shown in figure 3.1:

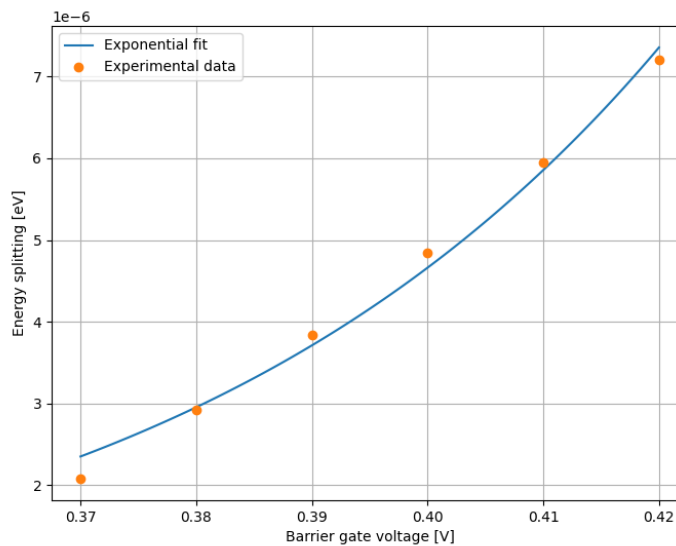


Figure 3.1: Plot of the fitted exponential compared with experimental data